

Explaining Legal Moves in Equations and Inequalities

Balance models, the properties of equality, and why one move flips an inequality sign

Directions

- A move is *legal* when it keeps the equation true. Doing the same thing to both sides is legal; rewriting one side into an equal expression is legal. Anything else is not.
 - When a problem asks you to name a move, say *what was done* and *that it was done to both sides* — for example, “subtracted $2x$ from both sides”.
 - Blanks inside the balance model are working space. The blank marked **Answer** is the graded one.
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1. A balance holds $x + 4$ grams on the left pan and 11 grams on the right pan, and it is level. Taking 4 grams off *both* pans keeps it level. Fill in the model to show the balance after that move, then give the value of x .

	Left pan	Right pan
Now	$x + 4$	11
After removing 4 from both pans	_____	_____

Answer: _____

2. Here is a complete solution, one line to the next:

$$3x - 7 = 14 \quad \longrightarrow \quad 3x = 21 \quad \longrightarrow \quad x = 7$$

(a) Name the move that turns the first line into the second.

Answer: _____

(b) Name the move that turns the second line into the third.

Answer: _____

(c) Use the same two kinds of move to solve $5x + 2 = 27$.

Answer: _____

3. Solve $4(x - 3) = 20$. Two different first moves are legal here: dividing both sides by 4, or rewriting the left side using the distributive property. Either one works.

Answer: _____

4. Solve the inequality $-3x < 12$. Write your answer as an inequality in x .

Answer: _____

5. A student solved an equation like this:

$$2x + 5 = 3x - 1 \quad \longrightarrow \quad 5 = x - 1 \quad \longrightarrow \quad 6 = x$$

- (a) Name the move used at each of the two steps.

Answer: _____

- (b) Solve the equation yourself and give the value of x .

Answer: _____

6. Solve $\frac{x}{5} - 3 = 4$.

Answer: _____

7. Consider the inequality $5 - 2x \geq 11$.

(a) Solve it. Write your answer as an inequality in x .

Answer: _____

(b) Say whether the inequality sign changes direction anywhere in your work, and name the exact step where that happens or explain why it never does.

Answer: _____

8. One of these steps is not legal:

$$3(x + 2) = 18 \longrightarrow 3x + 2 = 18 \longrightarrow 3x = 16 \longrightarrow x = \frac{16}{3}$$

(a) Name the step that is not legal and explain what rule it breaks.

Answer: _____

(b) Solve the equation correctly.

Answer: _____

9. Solve the inequality $-\frac{x}{4} + 1 > 3$. Write your answer as an inequality in x .

Answer: _____

10. Consider the equation $\frac{2x-1}{3} + 4 = x$.

(a) Solve it.

Answer: _____

(b) Explain which property justifies each move you made, and mark clearly which of your moves were done to *both* sides and which only rewrote *one* side.

Answer: _____